



Department of Statistics & Computer Science

University of Kelaniya

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COSC 21063 / BECS 21223 / COST 44233 / COST 44303

Data Structures and Algorithms

Practical Tutorial Guide 01

1. Write a program to reverse the following character sequence using the Stack operations.
 - **Input:** Enter your 1st name (example: Pasindu)
 - **Output:** udnisaP

2. Write a program to reverse the following number sequence using the Stack operations.
 - Input: 57693
 - Output: 39675

3. Write a Java program to read a sentence and reverse it word by word using a **linked implementation**. Display the reversed sentence as output.
 - Example: Input: Data Structures & Algorithm
 - Output: mhtiroglA & serutcurtS ataD

4. Write a Java program to determine whether a given word is a palindrome using stack operations.
 - HINT: A palindrome is a word that reads the same forward and backwards. If the word is a palindrome, display "Palindrome"; otherwise, display "Not a Palindrome".
 - Example:
Madam (Input): Palindrome.
Hello (Input): Not a Palindrome.

5. Write a program that reads an integer and prints it,

- a) Binary
- b) Octal
- c) Hexadecimal

Representation using the Stack operations. (Hint: Use stacks to store remainders while converting an integer to binary, octal, and hexadecimal.)

6. Create a small program to Stack Maximum Tracker using a **linked implementation**.

- a) Write a program that keeps track of the lowest value in a stack.
- b) Whenever a new element is pushed, update the current maximum if needed.
- c) After all elements are pushed, print the elements in ascending order.

Example:

- **Input sequence:** 5, 4, 9, 3, 10

Stepwise maximum update:

- Push 5 → Current Lowest Number: 5
- Push 4 → Current Lowest Number: 4
- Push 9 → Current Lowest Number: 4
- Push 3 → Current Lowest Number: 3
- Push 10 → Current Lowest Number: 3

- **Output:** 3, 4, 5, 9, 10

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